

## ANURAG KHANDLWAL

Associate Professor, Yale University

Email: [anurag.khandelwal@yale.edu](mailto:anurag.khandelwal@yale.edu)

Website: <https://www.anuragkhandelwal.com>

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### Education

**University of California Berkeley**

*PhD in Computer Science (2013 – 2019), Advised by Prof. Ion Stoica*

**Indian Institute of Technology, Kharagpur**

*B. Tech. in Computer Science and Engineering (2009 – 2013)*

### Work History

**Yale University, Department of Computer Science**

*Associate Professor, 2026 - Present*

*Assistant Professor, 2020 - 2026*

**Cornell University, Department of Computer Science (New York Campus)**

*Postdoctoral Associate, Fall 2019*

**Microsoft Research**

*Summer Research Intern, 2014 and 2012*

### Awards & Honors

**Best Paper Award, ACM ASPLOS 2026**

*CounterPoint: Using Hardware Event Counters to Refute & Refine Microarchitectural Assumptions*

**Best Paper Award Runner-Up, Best Artifact Award Runner-Up, ACM MobiSys 2026**

*TimelyLLM: Time-sensitive LLM Serving System for Physical-I/O Limited Agents*

**IEEE MICRO's Top-picks in Computer Architecture, 2025**

*CORD: Low-Latency, Bandwidth-Efficient and Scalable Release Consistency via Directory Ordering*

**Distinguished Artifact Award, ACM ISCA 2025**

*CORD: Low-Latency, Bandwidth-Efficient and Scalable Release Consistency via Directory Ordering*

**NetApp Faculty Fellowship, 2024**

*To develop GPU-native vector storage and indexing for accelerating AI workloads.*

**Best Student Paper Award, ACM EuroSys 2024**

*TRINITY: A Fast Compressed Multi-attribute Data Store*

**IEEE MICRO's Top-picks in Computer Architecture, 2023**

*SCALO: An Accelerator-Rich Distributed System for Scalable Brain-Computer Interfaces*

**Best Paper Award, ACM ISCA 2023**

*SCALO: An Accelerator-Rich Distributed System for Scalable Brain-Computer Interfaces*

**Roberts Innovation Award, 2023**

*For innovations in data center resource disaggregation.*

**NSF CAREER Award, 2021**

*For developing in-network memory management for disaggregated data centers.*

**NetApp Faculty Fellowship, 2021**

*For developing elastic and compressed disaggregated memory for edge data centers.*

**Distinguished Paper Award, USENIX Security 2020**

*For paper Pancake: Frequency Smoothing for Encrypted Data Stores*

**Madan Sundar Sahu Scholarship 2010, 2011, Indian Institute of Technology, Kharagpur**

*For outstanding academic performance during 2009-2010 and 2010-2011.*

## Research Focus

My research interests span computer software and hardware [systems](#), [networks](#), and [security](#). My work addresses challenges in processing, storing, and serving large volumes of data to empower real-world systems: from sprawling internet services like social media to critical tools in health and medicine.

## Research Impact

[2025] [CORD](#) incorporated in the design of [NVIDIA's Vera Rubin architecture](#).

[2024] [PromptCache](#) adopted in [Gemini](#), [OpenAI](#), and [Anthropic](#) for reusing attention states across LLM prompts.

[2024] [Trinity](#) adopted in NetApp's services to improve storage efficiency and performance.

[2023] The ideas underlying [SCALO](#) have been tested by Yale Neurosurgery in FPGA-based emulation frameworks for interfacing with the human brain in the operating room.

[2021] My [NSF CAREER Award project](#), [MIND](#), is the foundational system and central pillar of the multi-institution \$20-million NSF-funded AI Institute, [Athena](#).

[2019] [Confluo](#) employed in [Alibaba Cloud](#) for consuming and analyzing telemetry data.

[2015] [Succinct](#) employed by [Databricks customers](#) to query compressed RDDs.

## Publications

### Conferences & Workshops

- [C1] [Efficient and Scalable Synchronization via Generalized Cache Coherence](#)  
Yanpeng Yu, Seung-seob Lee, Lin Zhong, and Anurag Khandelwal  
*USENIX OSDI*, 2026
- [C2] [BulletTime: Time Dilation for High-Fidelity Tracing](#)  
Michael Wu, Sibren Isaacman, Abhishek Bhattacharjee, and Anurag Khandelwal  
*ACM ISCA*, 2026
- [C3] [TimelyLLM: Time-sensitive LLM Serving System for Physical-I/O Limited Agents](#)  
Neiwen Ling, Guojun Chen, Anurag Khandelwal, and Lin Zhong  
*ACM MobiSys*, 2026  
[Best Paper Award Runner-Up](#), [Best Artifact Award Runner-Up](#)
- [C4] [CounterPoint: Using Hardware Event Counters to Refute & Refine Microarchitectural Assumptions](#)  
Nick Lindsay, Caroline Trippel, Anurag Khandelwal, and Abhishek Bhattacharjee  
*ACM ASPLOS*, 2026  
[Best Paper Award](#)
- [C5] [Spirit: Fair Allocation of Interdependent Resources in Remote Memory Systems](#)  
Seung-seob Lee, Jachym Putta, Ziming Mao, and Anurag Khandelwal  
*ACM SOSP*, 2025
- [C6] [Scalable Far Memory: Balancing Faults and Evictions](#)  
Yueyang Pan\*, Yash Lala\*, Musa Unal, Yujie Ren, Seung-seob Lee, Abhishek Bhattacharjee, Anurag Khandelwal, and Sanidhya Kashyap  
*ACM SOSP*, 2025  
(\*Equal contribution authors)
- [C7] [Found in Translation: A Generative Language Modeling Approach to Memory Access Pattern Attacks](#)  
Grace Jia, Alex Wong, and Anurag Khandelwal  
*USENIX Security*, 2025
- [C8] [Weave: Efficient and Expressive Oblivious Analytics at Scale](#)  
Mahdi Soleimani, Grace Jia, and Anurag Khandelwal  
*USENIX OSDI*, 2025
- [C9] [CORD: Low-Latency, Bandwidth-Efficient and Scalable Release Consistency via Directory Ordering](#)  
Yanpeng Yu, Nicolai Oswald, and Anurag Khandelwal  
*ACM ISCA*, 2025  
[Distinguished Artifact Award](#), [IEEE Micro's Top Picks in Computer Architecture 2025](#)

- [C10] [PULSE: Accelerating Distributed Pointer-Traversals on Disaggregated Memory](#)  
Yupeng Tang, Seung-seob Lee, Abhishek Bhattacharjee, and Anurag Khandelwal  
*ACM ASPLOS*, 2025
- [C11] [Exploring Intelligent Dynamic Resource Provisioning for Elastic Massive MIMO vRAN](#)  
Parthiban Annamalai, Minsung Kim, Anurag Khandelwal, and Lin Zhong  
*ACM HotMobile*, 2025
- [C12] [Length Leakage in Oblivious Data Access Mechanisms](#)  
Grace Jia, Rachit Agarwal, and Anurag Khandelwal  
*USENIX Security*, 2024
- [C13] [Distributed Brain-Computer Interfacing With a Networked Multi-accelerator Architecture](#)  
Raghavendra Pothukuchi, Karthik Sriram, Michał Gerasimiuk, Muhammed Ugur, Rajit Manohar, Anurag Khandelwal, and Abhishek Bhattacharjee  
*IEEE Micro*, 2024
- [C14] [Prompt Cache: Modular attention reuse for low-latency inference](#)  
In Gim, Guojun Chen, Seung-seob Lee, Nikhil Sarda, Anurag Khandelwal, and Lin Zhong  
*MLSys*, 2024
- [C15] [Trinity: A Fast Compressed Multi-attribute Data Store](#)  
Ziming Mao, Kiran Srinivasan, and Anurag Khandelwal  
*ACM EuroSys*, 2024  
[Best Student Paper Award](#)
- [C16] [Karma: Resource Allocation for Dynamic Demands](#)  
Midhul Vuppapapati, Giannis Fikioris, Rachit Agarwal, Asaf Cidon, Anurag Khandelwal, and Éva Tardos  
*USENIX OSDI*, 2023
- [C17] [Prefetching Using Principles of Hippocampal-Neocortical Interaction](#)  
Michael Wu, Ketaki Joshi, Andrew Sheinberg, Guilherme Cox, Anurag Khandelwal, Raghavendra Pradyumna Pothukuchi, and Abhishek Bhattacharjee  
*ACM HotOS*, 2023
- [C18] [SCALO: An Accelerator-Rich Distributed System for Scalable Brain-Computer Interfacing](#)  
Karthik Sriram, Raghavendra Pothukuchi, Michał Gerasimiuk, Muhammed Ugur, Oliver Ye, Rajit Manohar, Anurag Khandelwal, and Abhishek Bhattacharjee  
*ACM ISCA*, 2023  
[Best Paper Award, IEEE Micro's Top Picks in Computer Architecture 2023](#)
- [C19] [Shepherd: Serving DNNs in the Wild](#)  
Hong Zhang, Yupeng Tang, Anurag Khandelwal, and Ion Stoica  
*USENIX NSDI*, 2023
- [C20] [Shortstack: Distributed, Fault-tolerant, Oblivious Data Access](#)  
Midhul Vuppapapati\*, Kushal Babel\*, Anurag Khandelwal, and Rachit Agarwal  
*USENIX OSDI*, 2022  
(\*Equal contribution authors)
- [C21] [Jiffy: Elastic Far-memory for Stateful Serverless Analytics](#)  
Anurag Khandelwal, Yupeng Tang, Rachit Agarwal, Aditya Akella, and Ion Stoica  
*ACM EuroSys*, 2022
- [C22] [MIND: In-Network Memory Management for Disaggregated Data Centers](#)  
Seung-seob Lee, Yanpeng Yu, Yupeng Tang, Anurag Khandelwal, Lin Zhong, and Abhishek Bhattacharjee  
*ACM SOSP*, 2021
- [C23] [What Serverless Computing Is and Should Become: The Next Phase of Cloud Computing](#)  
Johann Schleier-Smith, Vikram Sreekanti, Anurag Khandelwal, Joao Carreira, Neeraja J. Yadwadkar, Raluca Ada Popa, Joseph E. Gonzalez, Ion Stoica, and David A. Patterson  
*Communications of the ACM*, 2021
- [C24] [Caerus: Nimble Task Scheduling for Serverless Analytics](#)  
Hong Zhang, Yupeng Tang, Anurag Khandelwal, Jingrong Chen, and Ion Stoica  
*USENIX NSDI*, 2021

- [C25] [Pancake: Frequency Smoothing for Encrypted Data Stores](#)  
Paul Grubbs\*, Anurag Khandelwal\*, Marie-Sarah Lacharité\*, Lloyd Brown, Lucy Li, Rachit Agarwal, and Thomas Ristenpart  
*USENIX Security*, 2020  
(\*Equal contribution authors)  
[Distinguished Paper Award](#)
- [C26] [Le Taureau: Deconstructing the Serverless Landscape & A Look Forward](#)  
Anurag Khandelwal, Arun Kejariwal, and Karthikeyan Ramasamy  
*ACM SIGMOD*, 2020
- [C27] [Confluo: Distributed Monitoring and Diagnosis Stack for High-speed Networks](#)  
Anurag Khandelwal, Rachit Agarwal, and Ion Stoica  
*USENIX NSDI*, 2019
- [C28] [ZipG: A Memory-efficient Graph Store for Interactive Queries](#)  
Anurag Khandelwal, Zongheng Yang, Evan Ye, Rachit Agarwal, and Ion Stoica  
*ACM SIGMOD*, 2017
- [C29] [BlowFish: Dynamic Storage-Performance Tradeoff in Data Stores](#)  
Anurag Khandelwal, Rachit Agarwal, and Ion Stoica  
*USENIX NSDI*, 2016
- [C30] [Succinct: Enabling Queries on Compressed Data](#)  
Rachit Agarwal, Anurag Khandelwal, and Ion Stoica  
*USENIX NSDI*, 2015

## Theses

- [T1] [Queries on compressed data](#)  
Anurag Khandelwal

## Technical Reports

- [R1] [Cloud Programming Simplified: A Berkeley View on Serverless Computing](#)  
Eric Jonas, Johann Schleier-Smith, Vikram Sreekanti, Chia-Che Tsai, Anurag Khandelwal, Qifan Pu, Vaishaal Shankar, João Carreira, Karl Krauth, Neeraja J. Yadwadkar, Joseph E. Gonzalez, Raluca Ada Popa, Ion Stoica, and David A. Patterson  
*UC Berkeley Technical Report*, 2019
- [R2] [Attacking Data Center Networks from the Inside](#)  
Anurag Khandelwal, Navendu Jain, and Seny Kamara  
*Microsoft Research Technical Report*, 2015

## Preprints

- [P1] [Efficient MoE Serving in the Memory-Bound Regime: Balance Activated Experts, Not Tokens](#)  
Yanpeng Yu, Haiyue Ma, Krish Agarwal, Nicolai Oswald, Qijing Huang, Hugo Linsenmaier, Chunhui Mei, Ritchie Zhao, Ritika Borkar, Bitu Darvish Rouhani, David Nellans, Ronny Krashinsky, and Anurag Khandelwal  
*arXiv Paper 2512/09277*, 2025
- [P2] [Wiretapping LLMs: Network Side-Channel Attacks on Interactive LLM Services](#)  
Mahdi Soleimani, Grace Jia, In Gim, Seung-seob Lee, and Anurag Khandelwal  
*Cryptology ePrint Archive, Paper 2025/167*, 2025

## Invited Talks

### Performant Shared Disaggregated Memory

Carnegie Mellon University, November 2025  
University of Washington, October 2025  
University of Texas–Austin, October 2025  
Princeton University, October 2025  
University of Pennsylvania, October 2025  
Massachusetts Institute of Technology, September 2025  
New York University, March 2025  
Brown University, March 2025  
NSF AI Institute Showcase, Duke University, August 2024  
Microsoft Research, August 2024  
AMD Research, April 2024

### In-Network Memory Management for Disaggregated Data Centers

NSF AI Institute Showcase, Duke University, August 2022  
Cornell University, November 2021  
VMWare, November 2021  
Columbia University, October 2021

### Jiffy: Ephemeral Storage for Serverless Analytics

Facebook, March 2021  
Intel, March 2021

### Serverless streaming architectures & algorithms for the enterprise

Global STAC Live, October 2020 (*Industry conference for industry leaders in technology.*)  
Strata Data Conference, September 2019 (*Industry conference of over 5000 companies.*)

### Building Interactive Query Systems

Yale University, April 2019  
Purdue University, March 2019  
Microsoft Research, Redmond, March 2019  
Microsoft Research, Cambridge (UK), March 2019

### Realtime Monitoring and Analysis with Confluo

Splunk, San Francisco, February 2018

### Succinct: Enabling Queries on Compressed Data

SF Big Analytics Meetup, San Francisco, September 2016  
Bay Area AI Meetup, San Francisco, September 2016  
Cloudera, San Francisco, June 2016

## Funding

### NSF SaTC: Fortifying and Enriching Confidential Computing Environments

2025-28, \$900,000; PI: **Anurag Khandelwal**, Co-PIs: Lin Zhong, Seung-seob Lee

### NetApp Faculty Fellowship: AI-Native Vector Storage and Indexing

2024-25, \$117,686; PI: **Anurag Khandelwal**

### Yale Engineering AI Seed Grant: Brain-Inspired Memory Systems for AI Infrastructure

2024-25, \$150,000; PI: Abhishek Bhattacharjee, Co-PI: **Anurag Khandelwal**

### Roberts Innovation Fund: Disaggregated Edge Clouds

2023-24, \$50,000; PI: **Anurag Khandelwal**

### NSF RINGS: Intelligent and Resilient Virtualization of Massive MIMO Physical Layer

2022-25, \$1,000,000; PI: Lin Zhong, Co-PI: **Anurag Khandelwal**

### Swebilius Foundation Award: Systems for Brain Data Acquisition and Analytics over Multiple Modalities and Time-Scales

2022, \$16,000; PI: Abhishek Bhattacharjee, Co-PIs: **Anurag Khandelwal**, Rajit Manohar

NSF AI Institute: Edge Computing Leveraging Next Generation Networks

2021-26, \$20,000,000; Multi-institution, multi-PI grant, \$575,000 to **Khandelwal**

NSF PPOSS Planning: High-Performance Certified Trust for Global-Scale Applications

2021-22, \$250,000; PI: Zhong Shao, Co-PIs: Abhishek Bhattacharjee, **Anurag Khandelwal**, Lin Zhong

NSF SaTC: Mixed Distribution Models for Encrypted Data Stores

2021-25, \$1,000,000; PI: **Anurag Khandelwal**, Co-PIs: Rachit Agarwal, Thomas Ristenpart

NetApp Faculty Fellowship: A Unified Data Stack for Next-Generation IoT Systems

2021, \$50,000; PI: **Anurag Khandelwal**, Co-PI: Abhishek Bhattacharjee

NSF CAREER: In-Network Memory Management for Disaggregated Datacenters

2021-26, \$626,647.00; PI: **Anurag Khandelwal**

## Teaching

**CPSC422/522: Operating Systems**

Spring 2026: Course Director, 24 Lectures, Enrollment: 36 (YC)

Spring 2025: Course Director, 24 Lectures, Enrollment: 34 (YC)

Spring 2024: Course Director, 24 Lectures, Enrollment: 44 (YC)

**CPSC433/533: Computer Networks**

Spring 2022: Course Director, 24 Lectures, Enrollment: 39 (YC)

Spring 2021: Course Director, 24 Lectures, Enrollment: 34 (YC)

Spring 2020: Course Director, 24 Lectures, Enrollment: 44 (YC)

**CPSC 438/538: Big Data Systems**

Fall 2025: Course Director, 26 Lectures, Enrollment: 51 (YC)

Fall 2023: Course Director, 26 Lectures, Enrollment: 18 (YC)

Fall 2021: Course Director, 26 Lectures, Enrollment: 16 (YC)

**CPSC 638: Big Data Systems**

Fall 2020: Course Director, 26 Lectures, Enrollment: 3 (YC)

## Advising

**Postdoctoral Research:**

Seung-seob Lee (2020—, co-advised with Lin Zhong)

**Graduate Student Research:**

Grace Jia (2022—, PhD)

Mahdi Soleimani (2023—, PhD)

Yash Lala (2023—, PhD)

Stephen Chien (2026—, PhD)

Yanpeng Yu (2021—2026, PhD) ([→ Research Scientist at Databricks](#))

Yupeng Tang (2020—2024, PhD) ([→ Research Scientist at Meta Research](#))

Jachym Putta (2023—2025, Masters) ([→ TechOps Engineer at DE Shaw](#))

Leo Li (Masters) ([→ PhD at Yale](#))

**Undergraduate Student Research:** Rylan Yang (2025), Pax Ryan (2025), Rohan Thakur (2025—26),

Michael Tu (2024), Troy Feng (2023), Raymond Yang (2023),

Ziming Mao (2021—24) ([→ PhD at UC Berkeley](#)), Gabriel Buchdahl (2022—23),

Jachym Putta (2022—23), Adit Gupta (2022—23), Yuxuan Chen (2022),

Eli Sage-Martinson (2022), Stanley Wong (2022), Daniel Li (2021—22),

Brian Liu (2021), Alan Chen (2021), Jonathan Kraft (2021), Eric Chang (2020),

Jeff Ma ([→ PhD at University of Michigan](#))

## Education Outreach

**Yale Pathways to Science, Summer Scholars Program**

*"Through the Clouds: An Introduction to Cloud Computing"*

Instructor for 4-day workshop for high-school students (Summer 2021)

Instructor for 1-day seminar for high-school students (Summer 2022)

## Community Service

Organization Committee: CoNEXT'23 Poster Co-Chair  
Program Committee: SIGCOMM 2020, NSDI 2020, NSDI 2021, NSDI 2022, HotNets 2022, NSDI 2023, EuroSys 2023, SOSP 2024, EuroSys 2025, OSDI 2025, NSDI 2026, SOSP 2026, EuroSys 2027  
Journal Editorial Board (Serverless Area): JSys 2021  
External Program Committee: ASPLOS 2021  
NSF Panels: Undisclosed due to participant confidentiality.

## University Service

Yale Quantitative Reasoning Council: 2025-26  
Yale College Dean's Fellowship Review Committee: 2026, 2024

## Department Service

Chair, Colloquium Committee: Fall 2021  
CS Undergraduate Advisor: 2021-22 (Freshman/Sophomore), 2024-25 (Junior), 2025-26 (Junior)  
Yale CS Research Internship Program: Created and led the program (2021-25)  
Faculty Hiring Committee: 2025 (Database Systems Area Search)  
PhD Admissions Committee: 2020, 2021, 2022, 2023, 2024, 2025, 2026  
Summer Systems Seminars: 2020, 2021, 2022  
PhD Thesis Committee: Bo Hu (2021), Karthik Sriram (2023), Yupeng Tang (2024), Yanpeng Yu (2026), Nick Lindsay (2026)